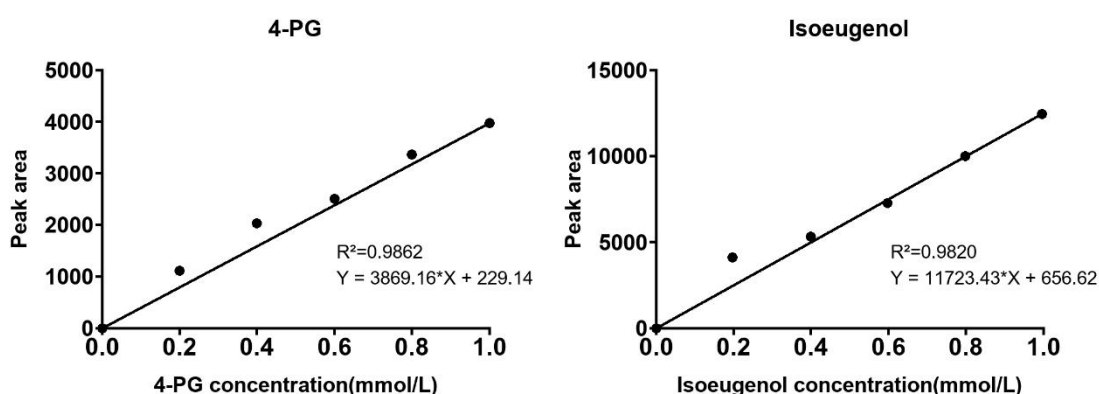


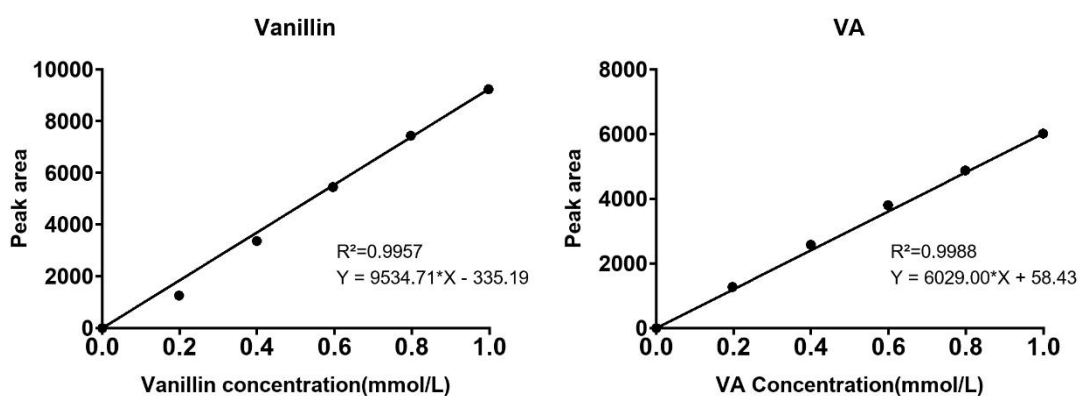
Supplementary Materials

Supplementary Table 1. Additional candidate genes mutations and their potential roles in tryptophan overproduction.

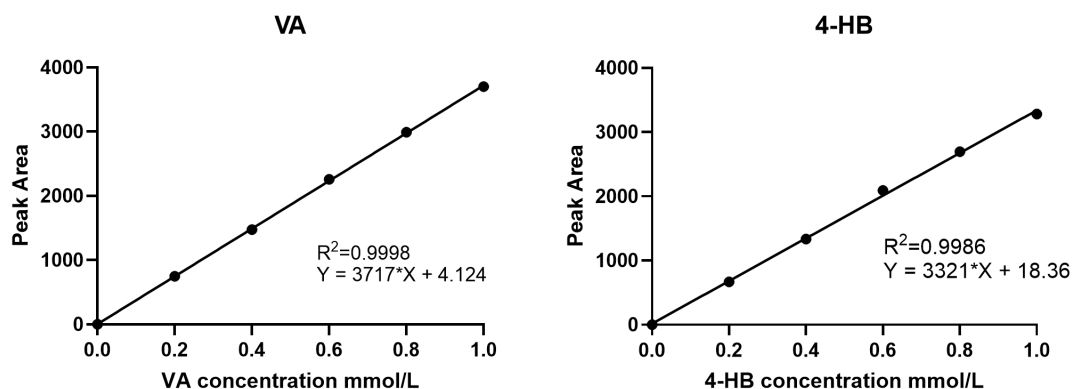
Strain	Functional Category	Mutation	Potential Role
1-20-7	Active pocket	16Glu→Gly	Shortens hydrogen bonds in the active pocket, optimizing substrate (VA) positioning
1-20-7	Active pocket	49Glu→Gly	Reshapes local residue flexibility, improving substrate entry and orientation
1-20-7	Substrate binding	107Arg→Cys	Modifies pocket charge environment, enhancing 4- HB/VA recognition and binding
1-20-7	Synonymous mutation	246Ala→Thr	Stabilizes local conformation without disrupting global folding



Supplementary Figure. The standard curve of 4-PG and Isoeugenol. (Module 1)



Supplementary Figure. The standard curve of vanillin and VA. (Module 1)



Supplementary Figure. The standard curve of VA and 4-HB. (Module 2)

Supplementary Table 2. Strains and plasmids used in this study.

Strains and plasmids	Description	Source
<i>E. coli</i> DH5α	F- <i>ΔlacU169</i> (Φ 80 <i>lacZ</i> Δ M15)	Invitrogen
<i>E. coli</i> BW25113	<i>hsdR17 recA1 endA1 supE44</i> F-, λ -, <i>E. coli</i> K-12 strain	Invitrogen
BW Δ <i>codA</i>	BD792 (CGSC6159) <i>lacZ</i>	Module 2
BW Δ <i>codA</i> -gRB2k- <i>pobR</i> ^{WT} -P _{pobA} *2- <i>mCherry-codA</i> -Cmr	BW25113 derivative, <i>ΔcodA</i> BW Δ <i>codA</i> carrying biosensor plasmid	Module 2
<i>E. coli</i> BW Δ <i>gldA</i> Δ <i>maeB</i> Δ <i>pykF</i>	gRB2k- <i>pobR</i> ^{WT} -P _{pobA} *2- <i>mCherry-codA</i> -Cmr, for ODM mutant screening BW25113 strain knocked out <i>gldA</i> , <i>maeB</i> and <i>pykF</i>	Module 3
Plasmids		
pYB1a- <i>progo</i>	<i>Amp</i> , <i>araC</i> , <i>progo</i>	Module 1
pYB1a- <i>novIf</i>	<i>Amp</i> , <i>araC</i> , <i>novIf</i>	Module 1
pYB1a- <i>vdh</i>	<i>Amp</i> , <i>araC</i> , <i>vdh</i>	Module 1
pYB1a- <i>nox</i>	<i>Amp</i> , <i>araC</i> , <i>nox</i>	Module 1
pYB1a- <i>progo-novIf</i>	<i>Amp</i> , <i>araC</i> , <i>progo</i> , <i>novIf</i>	Module 1
pYB1a- <i>vdh-nox</i>	<i>Amp</i> , <i>araC</i> , <i>vdh</i> , <i>nox</i>	Module 1
pRB1k- <i>vdh-nox</i>	<i>Kan</i> , <i>araC</i> , <i>vdh</i> , <i>nox</i>	Module 1

pYB1a- <i>ccdB</i>	Backbone vector for ODM mutant library construction, containing <i>ccdB</i> negative selection marker	Module 2
pYB1a- <i>omd</i> (wild-type)	pYB1a plasmid carrying wild-type <i>odm</i> gene for O-demethylase expression	Module 2
pYB1a- <i>omd</i> 1-20-7 (mutant library)	OMD mutant library generated by error-prone PCR and Gibson homologous recombination; library capacity 1.61×10 ⁵ clones	Module 2
pYB1a- <i>vanAB</i>	<i>Amp</i> , <i>araC</i> , <i>vanA</i> , <i>vanB</i>	Module 3
pLB1s- <i>vanAB-aroY-dmpD</i>	<i>Spe/Chl</i> , <i>araC</i> , <i>vanA</i> , <i>vanB</i> , <i>aroY</i> , <i>dmpD</i>	Module 3
pLB1s- <i>vanAB-aroY</i>	<i>Spe/Chl</i> , <i>araC</i> , <i>vanA</i> , <i>vanB</i> , <i>aroY</i>	Module 3
pLB1s- <i>aroY</i>	<i>Spe/Chl</i> , <i>araC</i> , <i>aroY</i>	Module 3
pYB1a- <i>dmpB-mhpD-mhpE</i>	<i>Amp</i> , <i>araC</i> , <i>dmpB</i> , <i>mhpD</i> , <i>mhpE</i>	Module 3
pYB1a- <i>dmpD</i>	<i>Amp</i> , <i>dmpD</i>	Module 3

Supplementary Table 3. Primers used in this study

pYB1a-<i>progo-novIf</i> construction	
PROGO-F	TAAGTGAATTCGGTGAGCTCGGTCTGC
PROGO-R	CACCTCCTTAGTACCAGATCTTTACAGATTCTGACCGCGAAAACGCTG
NOV1F-F	TAAAGATCTGGTACTAAGGAGGTGATGGCACAGTTTCCGAATACCCCG
NOV1F-R	GAGCTCACCGAATTCAGTTAGGCGGCCAGAACTTTTTCGGC
pYB1a-<i>vdh-nox</i> construction	
NOX-F-0811	CCGTTTAGTTAAAGATCTGGTACTAGTATGAGCAAGATCGTGATTGTTG
NOX-R-0811	CAGCTGCAGACCGAGCTCACCGAATCTTATTTTGCTTTCAGGGCGGC
pRB1k-<i>vdh-nox</i> construction	
KanR-F-0826	GATAAATGCTTCAATAATATTGAAAAAGGAAGAGTATGAGCCATATTCAACG

	GGAA
KanR-R-0826	CTAAAGTATATATGAGTAAACTTGGTC TGACAGTTATAAAAACTCATCGAGCA T
VN-F-0826	CTGTCAGACCAAGTTTACTCATATATA CTTTAGATTGATTTAAAACTTCATTTT T
VN-R-0826	ACTCTTCCTTTTTCAATATTATTGAAG CATTTATCAGGGTTATTGTCTCATGAG CG
VN-F-0904	GTACCTCTATGGCAAAAGTGAGCGCC GTTAAATATC
VN-R-0904	CTAGTACCAGATCTTTATTTTGCTTTC AGGGCGG
PR-F-0904	CAAAATAAAGATCTGGTACTAGTGGT GAATTC
PR-R-0904	CGGCGCTCACTTTTGCCATAGAGGTA CCCATGGTTAATTCCTCCTGTTAG
VN-F-0831	CTAACAGGAGGAATTAACCATGGGTA CCATGGCAAAAGTGAGCGCCG
VN-R-0831	ACCGAGCTCACC GAATTCACCACTAG TTTATTTTGCTTTCAGGGCGGC
pYB1a-<i>odm</i> construction	
OMD-error-F	GTGCTGGGTCTCGGATGATGAGTCT GATGGATAATTG
OMD-error-R	AGCGTGGGTCTCGGCATGTATTTGGT TTTCACTGTAC
Sequencing-R-0908	CTGCGTTCTGATTTAATCTG
gRB2k-<i>pobR</i>^{WT}-P_{pobA*2}-<i>mCherry-codA-Cmr</i> construction	
Vector-1-F	CAGCTATTCCGTTACTAGTGCATGCC ACACAC
Vector-1-R	CGTCTCATCGATGTTCACTGTGACGA G
Fragment-F	GAACGAGCTCCTTCCGCTTCCTCGCT CA
Fragment-R	CATGCACTAGTAACGGAATAGCTGTT CGTTG
Sequencing-F	GAGCAAAAAGCAAAGCACCG
Sequencing-R	CAGGAATCGAATGCAACCGG

Vector-2-F	CAGTGAACATCGATGAGACGGTCG
Vector-2-R	GAAGCGGAAGGAGCTCGTTCCTCCA ACGTTAAG
K-F	GCCACACACTCATTCAAATATGTATCC GTCATG
K-R	GTCTGACAGTTATAAAAACTCATCGA GCATCAAATGAAAC
pobR-1-F	CACCAATATCTAGCTCACCCAC
pobR-1-R	TATTTGAATGAGTGTGTGGCATGCAC TAG
pobR-2-F	AGTTTTTATAACTGTCAGACCAAGTT TACTCATATATACTTTAG
pobR-2-R	GTGGGTGAGCTAGATATTGGTG
pLB1s-<i>vanAB-aroY-dmpD</i> construction	
dmpD-F-0910	GTGAGCTCGGTCTGAGGATGAATGA ACGCACCG
dmpD-R-0910	CGGCACCAGCTGGCTAATGGCTGCT GCATCTG
PLB1S-aroY-0518-F	CATGACCGCTCCTATTCAGGATCTGC GTGATGCCATTGCC
PLB1S-aroY-0518-R	GTTAATTCCCTCCTGTTAGCCCGTTA GCCCCAAAAACGGG
pLB1s-F-0910	CATTAGCCAGCTGGTGCCGCGCGGC AG
pLB1s-R-0910	CACTCCTCAGACCGAGCTCACCGAAT TC
VanA-F-0511	GGGCTAACAGGAGGGAATTAACCAT GGCATGGGTACAGAAACCGTTGC
VanB-R-0511	CCTGAATAGGAGCGGTCATGGTACCC TCCTCTTACAGATCCAGCACCAAGTC
pYB1a-<i>dmpB-mhpD-mhpE</i> construction	
dmpB-F-0829	CATACCCGTTTTTTGGGCTAACNNNN NGAATTAACC
dmpB-R-0829	AGTAGATCTTTAGGTCAGCACGGTCA GAAAACG
mhpD-F-0829	CATTAGCGGTACTAGTACTNNNNNTG AATGACG
mhpD-R-0829	AGTGAATTCACCTCATGACAGACTTC CTTTTG

mhpE-F-0829	GTCATGAGGTGAATTCACTNNNNNTG AGG
mhpE-R-0829	GTTAGCCCCAAAAAACGGGTATGGA GAAACGTTAGCC
pYB1a-<i>dmpD</i> construction	
dmpD-F-0829	AGTAGATCTTTAGGTCAGCACGGTCA GAAAACG
dmpD-F-0902	CTGACCTAAAGATCTACTNNNNNTGA ATGAACGCACCGCAGAATAGTCCG
dmpD-R-0829	AGTACTAGTACCGCTAATGGCTGCTG CATCTG
